

iC-MB4

BiSS Interface Master

Description

iC-MB4 is a BiSS/SSI interface bridge component suitable to implement BiSS/SSI master functionality into any device or platform.

BiSS C, BiSS B, and SSI slave devices (e.g. sensors or actuators) can be fully accessed by iC-MB4. Up to 8 devices can be connected, each device with up to 64 bit single-cycle data length for sensor and actuator data and up to 16 bit CRC polynomial.

Two physical BiSS/SSI channels enable single-ended communication via internal CMOS and TTL transceivers. Alternatively, one differential BiSS/SSI interface with on-chip RS422 and LVDS transceivers can be enabled.

iC-MB4 is configured and controlled by a host microcontroller via an 8-bit parallel or serial interface (SPI). The sensor/actuator data transfer is started by a microcontroller instruction, via pin GETSENS or by iC-MB4 itself using automatic sensor data acquisition.

The sensor/actuator data transmission end and status are indicated at pins EOT and NER. Status registers provide details on successful single-cycle data and register communication. An internal 20 MHz oscillator generates the system clock frequency. Alternatively, a high-precision external oscillator can be applied.

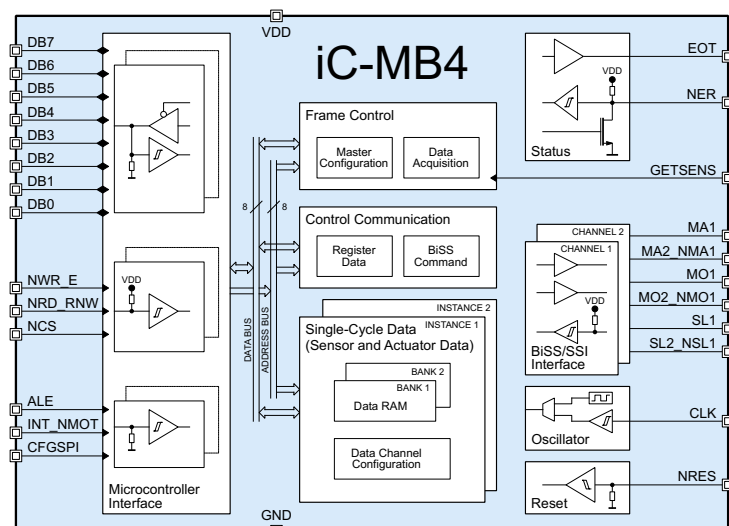
Features

- Bidirectional BiSS communication with up to 8 slaves
- Supports SSI protocol for unidirectional data transmission
- Synchronous and cyclic sensor data acquisition with data rates up to 10 Mbit/s
- Configurable BiSS/SSI interface: two single-ended channels with TTL/CMOS transceivers or one differential channel with RS422/LVDS transceivers
- Slave register operations during cyclic data transfers
- Automatic line delay compensation
- Configurable data lengths of up to 64 bit for each sensor
- CRC verification with up to 16 bits per slave
- Separate memory banks for free controller access during BiSS sensor data transfers
- 64 bytes memory for bidirectional register access
- Parallel or serial (SPI) interface to host e.g. MCU
- Single 3 V to 5 V supply

Applications

- Bidirectional communication in multi sensor systems
- Linear and rotary encoders
- Motor feedback systems
- PLCs
- Drives
- Robotics

Block Diagram



BiSS Interface Master

Key Specifications

General

Supply Voltage Input VDD	3V to 5.5V
Supply Current	Max. 20 mA (no load)
Operating Temperature	−40°C to +85 °C −40°C to +125 °C (optional)
Package (RoHS compliant)	28-pin QFN 5.0 mm × 5.0 mm, thickness 0.9 mm
On-Chip System Clock Oscillator	20 MHz

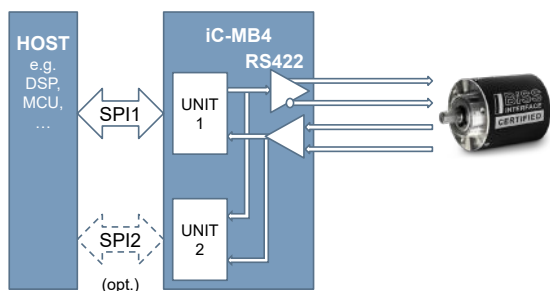
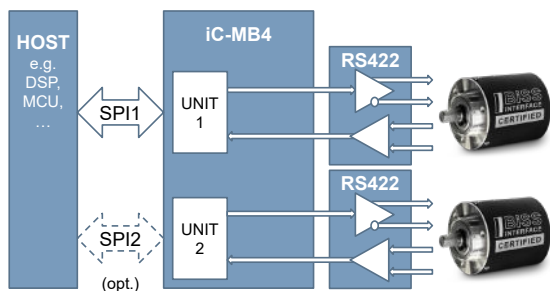
Microcontroller Interface

Parallel Interface	Motorola 68HC11, Intel 8051
Serial Interface	10 MHz SPI, second SPI for redundant single-cycle data access

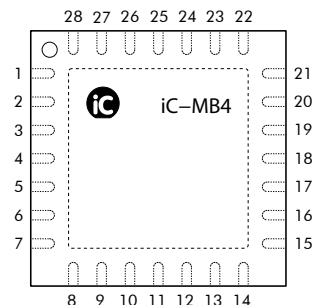
BiSS/SSI Interface

Protocols	BiSS C, BiSS B, SSI
Transmission Rate	Up to 10 Mbit/s
Slave Count	Up to 8 slaves
Clock Frequency Range	62.5 kHz to 10 MHz
Frame Repetition Rate	1 μ s to 4 ms
Transceivers	TTL/CMOS for single-ended operation (up to 2 physical channels), RS422/LVDS for differential operation (1 physical channel)

Application Examples

Differential operation using **on-chip** RS422 transceivers (**1 physical channel**)Differential operation using **external** RS422 transceivers (**2 physical channels**)

Pin Configuration QFN28-5×5



Pin Functions

No.	Name	Function
Serial Microcontroller Interface (SPI)		
27	NCS	SPI Chip Select Input (NCS) (active low)
28	ALE	SPI Clock Input (SCLK)
1	DB0	SPI Serial Data Input (MOSI)
2	DB1	SPI Serial Data Output (MISO)
5	DB4	SPI2 Chip Select Input (NCS) (active low)
6	DB5	SPI2 Clock Input (SCLK)
7	DB6	SPI2 Serial Data Input (MOSI)
8	DB7	SPI2 Serial Data Output (MISO)
Parallel Microcontroller Interface: Intel Mode		
25	NWR_E	Write Input (active low)
26	NRD_RNW	Read Input (active low)
27	NCS	Chip Select Input (active low)
28	ALE	Address Latch Enable Input
1...8	DB0...DB7	Data Bus Input/Output
Parallel Microcontroller Interface: Motorola Mode		
25	NWR_E	Enable Input (active high)
26	NRD_RNW	Read/Not-Write Select Input
27	NCS	Chip Select Input (active low)
28	ALE	Address Latch Enable Input
1...8	DB0...DB7	Data Bus Input/Output
Supply Voltage		
9	GND	Ground
10	VDD	+3V . . . +5.5V Supply Voltage
Status and Frame Control		
11	EOT	End Of Transmission Output
12	GETSENS	Sensor Data Request Input
13	NER	Error Input/Output (active low)
14	n.c.	not connected
BiSS/SSI Interface		
15	MA1	BiSS Clock Line Output
16	MA2_NMA1	BiSS Clock Line Output Ch. 2
17	SL1	BiSS Data Line Input
18	SL2_NSL1	BiSS Data Line Input Ch. 2
23	MO1	BiSS Data Line Output
24	MO2_NMO1	BiSS Data Line Output Ch. 2
Configuration, Reset and Oscillator		
19	INT_NMOT	Communication Mode Select Input (Intel = 1, Motorola = 0)
20	CFGSPI	MCU Interface Select Input (serial (SPI) = 1, parallel = 0)
21	NRES	Reset Input (active low)
22	CLK	External Clock Input